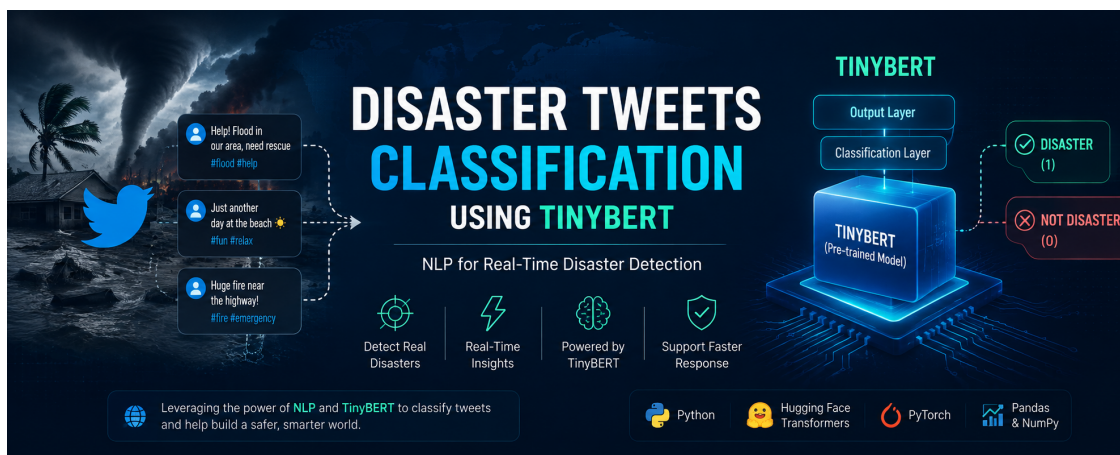


Disaster_Tweets_Classification_Using_TinyBERT

August 6, 2026

1 Disaster Tweets Classification Using TinyBERT



1.1 Download the Data

```
[1]: import pandas as pd
import warnings
warnings.filterwarnings("ignore")
```

```
[2]: df = pd.read_csv("https://raw.githubusercontent.com/laxmimerit/
↳All-CSV-ML-Data-Files-Download/refs/heads/master/twitter_disaster_tweets.
↳csv", usecols=['text', 'target'])
df.head()
```

```
[2]:
```

	text	target
0	Our Deeds are the Reason of this #earthquake M...	1
1	Forest fire near La Ronge Sask. Canada	1
2	All residents asked to 'shelter in place' are ...	1
3	13,000 people receive #wildfires evacuation or...	1
4	Just got sent this photo from Ruby #Alaska as ...	1

```
[3]: df.isnull().sum()
```

```
[3]: text      0
target      0
```

dtype: int64

```
[4]: df = df.sample(frac=1).reset_index(drop=True)
df.head()
```

```
[4]:
```

	text	target
0	Listen to Landslide by Oh Wonder #SoundCloud h...	0
1	@LayLoveTournay @RyroTheUnaware [loud groaning...	0
2	the apocalypse is upon us	0
3	DRob collided into Dan Hughes while she was go...	0
4	I'm a disaster?? https://t.co/VCV73BUaCZ	0

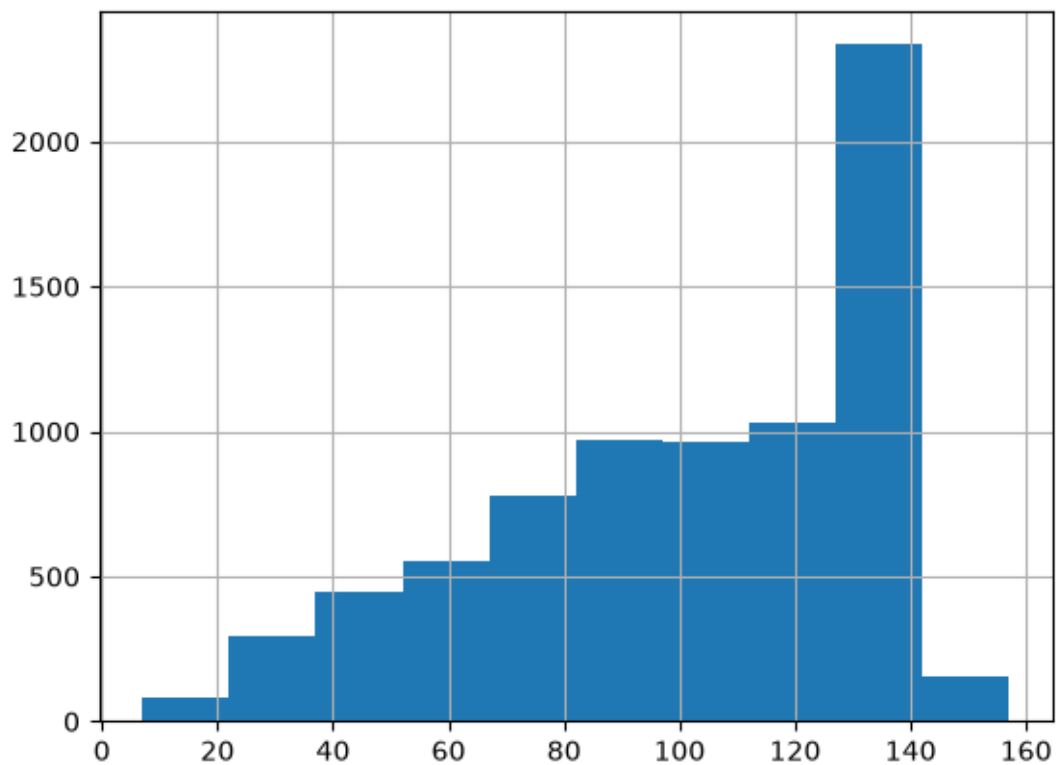
```
[5]: df['target'].value_counts()
```

```
[5]: target
0    4342
1     3271
Name: count, dtype: int64
```

```
[6]: # context window in transformer -> 512 token
# 1 token -> 3 char

df['text'].str.len().hist()
```

```
[6]: <Axes: >
```



2 Load Data with Hugging Face Datasets Library

```
[7]: df = df.rename(columns={'target': 'label'})
```

```
[8]: from datasets import Dataset
dataset = Dataset.from_pandas(df)
dataset = dataset.train_test_split(test_size=0.2)
```

```
[9]: dataset
```

```
[9]: DatasetDict({
  train: Dataset({
    features: ['text', 'label'],
    num_rows: 6090
  })
  test: Dataset({
    features: ['text', 'label'],
    num_rows: 1523
  })
})
```

```
[10]: dataset['train'][0]
```

```
[10]: {'text': 'Washington Post - 4 dead dozens injured in Gaza blast near house
leveled in summer war http://t.co/VjXa13n8Ap',
  'label': 1}
```

```
[11]: id2label = {0: 'general', 1: 'disaster'}
label2id = {'general': 0, 'disaster': 1}
```

3 Data Tokenization

```
[12]: from transformers import AutoTokenizer
import torch

device = torch.device('cuda' if torch.cuda.is_available() else torch.
↳ device('cpu'))

model_ckpt = 'huawei-noah/TinyBERT_General_4L_312D'
tokenizer = AutoTokenizer.from_pretrained(model_ckpt, use_fast = True)
```

```
[ ]:
```

```
[13]: def tokenize(batch):
        temp = tokenizer(batch['text'], padding=True, truncation =
        ↪ True,max_length=100)
        return temp
dataset = dataset.map(tokenize, batched=True, batch_size = None)
```

```
Map: 100%|      | 6090/6090 [00:01<00:00, 4974.96 examples/s]
Map: 100%|      | 1523/1523 [00:00<00:00, 3497.71 examples/s]
```

```
[14]: dataset
```

```
[14]: DatasetDict({
    train: Dataset({
        features: ['text', 'label', 'input_ids', 'token_type_ids',
'attention_mask'],
        num_rows: 6090
    })
    test: Dataset({
        features: ['text', 'label', 'input_ids', 'token_type_ids',
'attention_mask'],
        num_rows: 1523
    })
})
```

4 Building Model Evaluation Function

```
[15]: !pip install evaluate
      !pip install sequeval
```

```
Requirement already satisfied: evaluate in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (0.4.6)
Requirement already satisfied: datasets>=2.0.0 in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from evaluate) (5.0.1)
Requirement already satisfied: numpy>=1.17 in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from evaluate) (1.26.4)
Requirement already satisfied: dill in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from evaluate) (0.4.1)
Requirement already satisfied: pandas in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from evaluate) (3.0.3)
Requirement already satisfied: requests>=2.19.0 in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from evaluate) (2.33.1)
Requirement already satisfied: tqdm>=4.62.1 in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from evaluate) (4.68.2)
Requirement already satisfied: xxhash in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from evaluate) (3.7.0)
Requirement already satisfied: multiprocessing in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from evaluate) (0.70.19)
Requirement already satisfied: fsspec>=2021.05.0 in
```

C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
 fsspec[http]>=2021.05.0->evaluate) (2026.4.0)
 Requirement already satisfied: huggingface-hub>=0.7.0 in
 C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from evaluate) (0.36.2)
 Requirement already satisfied: packaging in
 C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from evaluate) (24.2)
 Requirement already satisfied: filelock in
 C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
 datasets>=2.0.0->evaluate) (3.29.1)
 Requirement already satisfied: pyarrow>=21.0.0 in
 C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
 datasets>=2.0.0->evaluate) (23.0.1)
 Requirement already satisfied: httpx<1.0.0 in
 C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
 datasets>=2.0.0->evaluate) (0.28.1)
 Requirement already satisfied: pyyaml>=5.1 in
 C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
 datasets>=2.0.0->evaluate) (6.0.3)
 Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in
 C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
 fsspec[http]>=2021.05.0->evaluate) (3.14.1)
 Requirement already satisfied: anyio in
 C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
 httpx<1.0.0->datasets>=2.0.0->evaluate) (4.14.1)
 Requirement already satisfied: certifi in
 C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
 httpx<1.0.0->datasets>=2.0.0->evaluate) (2026.6.17)
 Requirement already satisfied: httpcore==1.* in
 C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
 httpx<1.0.0->datasets>=2.0.0->evaluate) (1.0.9)
 Requirement already satisfied: idna in
 C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
 httpx<1.0.0->datasets>=2.0.0->evaluate) (3.18)
 Requirement already satisfied: h11>=0.16 in
 C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
 httpcore==1.*->httpx<1.0.0->datasets>=2.0.0->evaluate) (0.16.0)
 Requirement already satisfied: typing-extensions>=3.7.4.3 in
 C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from huggingface-
 hub>=0.7.0->evaluate) (4.15.0)
 Requirement already satisfied: aiohappyeyeballs>=2.5.0 in
 C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
 aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (2.6.2)
 Requirement already satisfied: aiosignal>=1.4.0 in
 C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
 aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (1.4.0)
 Requirement already satisfied: attrs>=17.3.0 in
 C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
 aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (26.1.0)

```

Requirement already satisfied: frozenlist>=1.1.1 in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (6.7.1)
Requirement already satisfied: propcache>=0.2.0 in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (0.5.2)
Requirement already satisfied: yarl<2.0,>=1.17.0 in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (1.24.2)
Requirement already satisfied: charset_normalizer<4,>=2 in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
requests>=2.19.0->evaluate) (3.4.7)
Requirement already satisfied: urllib3<3,>=1.26 in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
requests>=2.19.0->evaluate) (2.7.0)
Requirement already satisfied: colorama in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from
tqdm>=4.62.1->evaluate) (0.4.6)
Requirement already satisfied: python-dateutil>=2.8.2 in
C:\Users\Uditya\AppData\Roaming\Python\Python311\site-packages (from
pandas->evaluate) (2.9.0.post0)
Requirement already satisfied: tzdata in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from pandas->evaluate)
(2026.2)
Requirement already satisfied: six>=1.5 in
C:\Users\Uditya\AppData\Roaming\Python\Python311\site-packages (from python-
dateutil>=2.8.2->pandas->evaluate) (1.17.0)
Collecting sequeval
  Downloading sequeval-1.2.2.tar.gz (43 kB)
  Installing build dependencies: started
  Installing build dependencies: finished with status 'done'
  Getting requirements to build wheel: started
  Getting requirements to build wheel: finished with status 'done'
  Installing backend dependencies: started
  Installing backend dependencies: finished with status 'done'
  Preparing metadata (pyproject.toml): started
  Preparing metadata (pyproject.toml): finished with status 'done'
Requirement already satisfied: numpy>=1.14.0 in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from sequeval) (1.26.4)
Requirement already satisfied: scikit-learn>=0.21.3 in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from sequeval) (1.9.0)
Requirement already satisfied: scipy>=1.10.0 in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from scikit-
learn>=0.21.3->sequeval) (1.17.1)
Requirement already satisfied: joblib>=1.4.0 in

```

```

C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from scikit-learn>=0.21.3->sequeval) (1.5.3)
Requirement already satisfied: narwhals>=2.0.1 in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from scikit-learn>=0.21.3->sequeval) (2.22.1)
Requirement already satisfied: threadpoolctl>=3.5.0 in
C:\Users\Uditya\anaconda3\envs\genai\Lib\site-packages (from scikit-learn>=0.21.3->sequeval) (3.6.0)
Building wheels for collected packages: sequeval
  Building wheel for sequeval (pyproject.toml): started
  Building wheel for sequeval (pyproject.toml): finished with status 'done'
  Created wheel for sequeval: filename=sequeval-1.2.2-py3-none-any.whl size=16281 sha256=c0a494a070587b1b7eb7b991fda9fc6f8be5903aab70f40cf0bffb2817a3cd7c
  Stored in directory: c:\users\uditya\appdata\local\pip\cache\wheels\bc\92\f0\243288f899c2eacdfa8c5f9aede4c71a9bad0ee26a01dc5ead
Successfully built sequeval
Installing collected packages: sequeval
Successfully installed sequeval-1.2.2

```

```

[16]: import evaluate
import numpy as np

accuracy = evaluate.load("accuracy")

def compute_metrics(eval_pred):
    predictions, labels = eval_pred
    predictions = np.argmax(predictions, axis=1)

    return accuracy.compute(
        predictions=predictions,
        references=labels
    )

```

5 Model Building and Training

```

[17]: from transformers import AutoModelForSequenceClassification, TrainingArguments, \
    Trainer

model = AutoModelForSequenceClassification.from_pretrained(
    model_ckpt,
    num_labels=len(label2id),
    id2label=id2label,
    label2id=label2id,
)

```

Some weights of BertForSequenceClassification were not initialized from the model checkpoint at huawei-noah/TinyBERT_General_4L_312D and are newly

```
initialized: ['classifier.bias', 'classifier.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.
```

```
[18]: model
```

```
[18]: BertForSequenceClassification(
  (bert): BertModel(
    (embeddings): BertEmbeddings(
      (word_embeddings): Embedding(30522, 312, padding_idx=0)
      (position_embeddings): Embedding(512, 312)
      (token_type_embeddings): Embedding(2, 312)
      (LayerNorm): LayerNorm((312,), eps=1e-12, elementwise_affine=True)
      (dropout): Dropout(p=0.1, inplace=False)
    )
    (encoder): BertEncoder(
      (layer): ModuleList(
        (0-3): 4 x BertLayer(
          (attention): BertAttention(
            (self): BertSdpaSelfAttention(
              (query): Linear(in_features=312, out_features=312, bias=True)
              (key): Linear(in_features=312, out_features=312, bias=True)
              (value): Linear(in_features=312, out_features=312, bias=True)
              (dropout): Dropout(p=0.1, inplace=False)
            )
            (output): BertSelfOutput(
              (dense): Linear(in_features=312, out_features=312, bias=True)
              (LayerNorm): LayerNorm((312,), eps=1e-12, elementwise_affine=True)
              (dropout): Dropout(p=0.1, inplace=False)
            )
          )
          (intermediate): BertIntermediate(
            (dense): Linear(in_features=312, out_features=1200, bias=True)
            (intermediate_act_fn): GELUActivation()
          )
          (output): BertOutput(
            (dense): Linear(in_features=1200, out_features=312, bias=True)
            (LayerNorm): LayerNorm((312,), eps=1e-12, elementwise_affine=True)
            (dropout): Dropout(p=0.1, inplace=False)
          )
        )
      )
    )
    (pooler): BertPooler(
      (dense): Linear(in_features=312, out_features=312, bias=True)
      (activation): Tanh()
    )
  )
)
```

```
)
(dropout): Dropout(p=0.1, inplace=False)
(classifier): Linear(in_features=312, out_features=2, bias=True)
)
```

```
[ ]: from transformers import TrainingArguments, Trainer # Added Trainer here
```

```
args = TrainingArguments(
    output_dir="out",
    logging_dir="logs",
    report_to="none",
    logging_strategy="steps",
    logging_steps=50,

    num_train_epochs=3,
    learning_rate=2e-5,
    per_device_train_batch_size=32,
    per_device_eval_batch_size=32,
    eval_strategy="epoch", # Changed from evaluation_strategy
    save_strategy="epoch",
)

trainer = Trainer(
    model=model,
    args=args,
    train_dataset=dataset["train"],
    eval_dataset=dataset["test"],
    compute_metrics=compute_metrics,
    tokenizer=tokenizer,
)
```

```
[20]: trainer.train()
```

```
9%|          | 50/573 [00:33<04:48, 1.82it/s]
{'loss': 0.6637, 'grad_norm': 2.0786540508270264, 'learning_rate':
1.825479930191972e-05, 'epoch': 0.26}

17%|         | 100/573 [01:00<04:10, 1.89it/s]
{'loss': 0.5587, 'grad_norm': 2.2530975341796875, 'learning_rate':
1.6509598603839444e-05, 'epoch': 0.52}

26%|         | 150/573 [01:28<03:37, 1.94it/s]
{'loss': 0.4948, 'grad_norm': 9.96387767791748, 'learning_rate':
1.4764397905759162e-05, 'epoch': 0.79}

33%|         | 191/573 [01:54<02:39, 2.39it/s]
```

```

{'eval_loss': 0.4526946246623993, 'eval_accuracy': 0.81483913328956,
'eval_runtime': 5.7794, 'eval_samples_per_second': 263.524,
'eval_steps_per_second': 8.305, 'epoch': 1.0}

35%|          | 200/573 [01:59<03:48, 1.63it/s]

{'loss': 0.4666, 'grad_norm': 5.314987659454346, 'learning_rate':
1.3019197207678885e-05, 'epoch': 1.05}

44%|          | 250/573 [02:25<02:42, 1.99it/s]

{'loss': 0.448, 'grad_norm': 4.08004903793335, 'learning_rate':
1.1273996509598604e-05, 'epoch': 1.31}

52%|          | 300/573 [02:50<02:17, 1.98it/s]

{'loss': 0.4497, 'grad_norm': 17.55708122253418, 'learning_rate':
9.528795811518325e-06, 'epoch': 1.57}

61%|          | 350/573 [03:16<01:52, 1.97it/s]

{'loss': 0.4397, 'grad_norm': 4.268263339996338, 'learning_rate':
7.783595113438046e-06, 'epoch': 1.83}


67%|          | 382/573 [03:40<01:27, 2.18it/s]

{'eval_loss': 0.42150840163230896, 'eval_accuracy': 0.8214051214707814,
'eval_runtime': 7.0082, 'eval_samples_per_second': 217.317,
'eval_steps_per_second': 6.849, 'epoch': 2.0}

70%|          | 400/573 [03:51<01:34, 1.82it/s]

{'loss': 0.431, 'grad_norm': 5.231829643249512, 'learning_rate':
6.038394415357767e-06, 'epoch': 2.09}

79%|          | 450/573 [04:17<01:01, 1.99it/s]

{'loss': 0.4156, 'grad_norm': 5.330288887023926, 'learning_rate':
4.293193717277487e-06, 'epoch': 2.36}

87%|          | 500/573 [04:44<00:42, 1.73it/s]

{'loss': 0.3964, 'grad_norm': 5.9636335372924805, 'learning_rate':
2.547993019197208e-06, 'epoch': 2.62}

96%|          | 550/573 [05:09<00:11, 1.98it/s]

{'loss': 0.4192, 'grad_norm': 6.270261764526367, 'learning_rate':
8.027923211169285e-07, 'epoch': 2.88}


100%|         | 573/573 [05:28<00:00, 2.39it/s]

{'eval_loss': 0.42383870482444763, 'eval_accuracy': 0.8194353250164149,
'eval_runtime': 6.055, 'eval_samples_per_second': 251.526,
'eval_steps_per_second': 7.927, 'epoch': 3.0}

```

```
100%|      | 573/573 [05:29<00:00, 1.74it/s]
```

```
{'train_runtime': 329.5304, 'train_samples_per_second': 55.443,  
'train_steps_per_second': 1.739, 'train_loss': 0.46924917943398575, 'epoch':  
3.0}
```

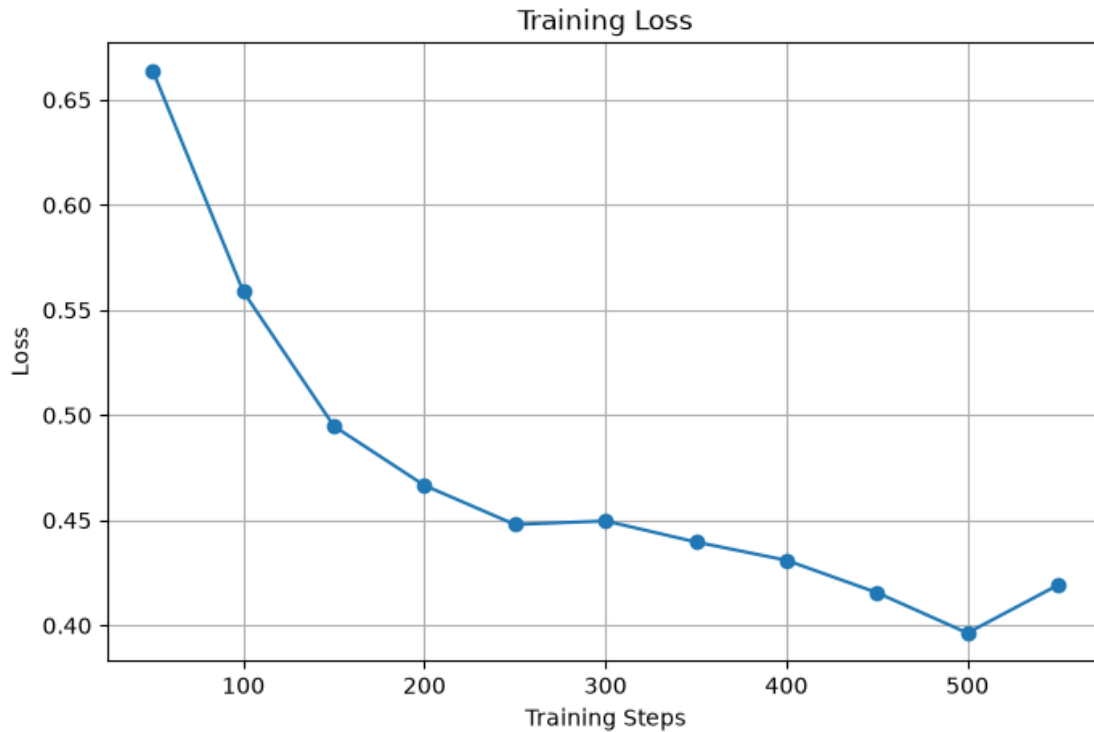
```
[20]: TrainOutput(global_step=573, training_loss=0.46924917943398575,  
metrics={'train_runtime': 329.5304, 'train_samples_per_second': 55.443,  
'train_steps_per_second': 1.739, 'total_flos': 42980020947360.0, 'train_loss':  
0.46924917943398575, 'epoch': 3.0})
```

```
[21]: trainer.evaluate()
```

```
100%|      | 48/48 [00:05<00:00, 8.25it/s]
```

```
[21]: {'eval_loss': 0.42383870482444763,  
'eval_accuracy': 0.8194353250164149,  
'eval_runtime': 6.3565,  
'eval_samples_per_second': 239.598,  
'eval_steps_per_second': 7.551,  
'epoch': 3.0}
```

```
[22]: import matplotlib.pyplot as plt  
  
logs = trainer.state.log_history  
  
train_loss = []  
steps = []  
  
for log in logs:  
    if "loss" in log:  
        train_loss.append(log["loss"])  
        steps.append(log["step"])  
  
plt.figure(figsize=(8,5))  
plt.plot(steps, train_loss, marker="o")  
plt.title("Training Loss")  
plt.xlabel("Training Steps")  
plt.ylabel("Loss")  
plt.grid(True)  
plt.show()
```

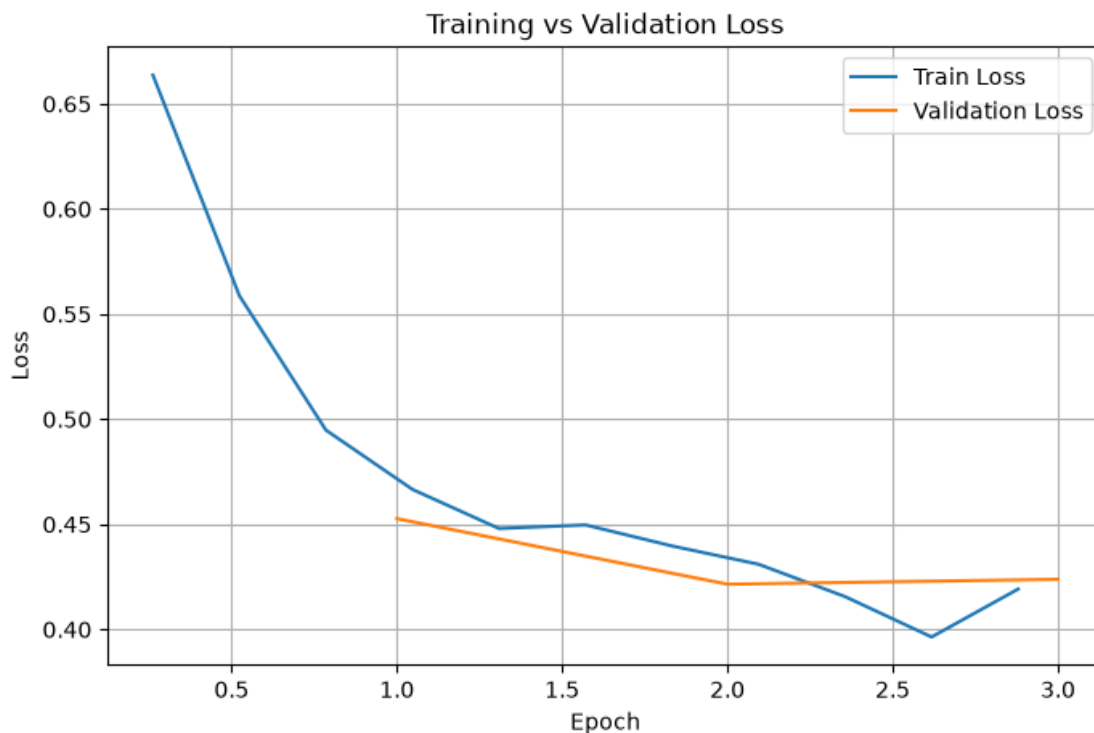


```
[23]: train_epochs = []
train_loss = []
eval_epochs = []
eval_loss = []

for log in trainer.state.log_history:
    if "loss" in log and "epoch" in log:
        train_epochs.append(log["epoch"])
        train_loss.append(log["loss"])

    if "eval_loss" in log:
        eval_epochs.append(log["epoch"])
        eval_loss.append(log["eval_loss"])

plt.figure(figsize=(8,5))
plt.plot(train_epochs, train_loss, label="Train Loss")
plt.plot(eval_epochs, eval_loss, label="Validation Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Training vs Validation Loss")
plt.legend()
plt.grid(True)
plt.show()
```



6 Model Saving and Load for Inference

```
[24]: trainer.save_model('tinybert-disaster-tweet')
```

```
[10]: from transformers import pipeline
import torch
text = [
    "BREAKING: A massive 6.8 magnitude earthquake just struck off the coast.␣
    ↳Tsunami warnings issued for coastal areas! #Earthquake #News",
    "My laptop just crashed right before I saved my 20-page term paper. This is␣
    ↳a total disaster #fml #help",
    "The wildfire is spreading rapidly due to high winds. Evacuation orders are␣
    ↳now in place for Northern County residents. Be safe!",
    "That new Taylor Swift track is absolute fire! I have had it on repeat all␣
    ↳morning long ",
    "Flash flood warnings active for downtown. Multiple roads are completely␣
    ↳submerged. Do not attempt to drive through flooded areas.",
    "We are drowning in paperwork at the office today. Someone please rescue me␣
    ↳from this boring meeting.",
    "Emergency crews are responding to a structural collapse at the industrial␣
    ↳park. Avoid the area to let sirens through.",
```

```

    "I am burning up in this summer heat! The AC at the gym is broken and it
    feels like a literal oven in here.",
    "A major multi-vehicle accident has completely blocked all southbound lanes
    on Interstate 95. Expect severe traffic delays.",
    "My favorite character just got killed off in the season finale. I am
    emotionally destroyed right now. Why did they do this?!"
]

device = torch.device('cuda') if torch.cuda.is_available() else torch.
device('cpu')

classifier = pipeline('text-classification', model = 'tinybert-disaster-tweet',
device = device)

df = classifier(text)

```

```
[11]: df
```

```

[11]: [{'label': 'disaster', 'score': 0.9027481079101562},
      {'label': 'general', 'score': 0.7186434268951416},
      {'label': 'disaster', 'score': 0.9023825526237488},
      {'label': 'general', 'score': 0.8390481472015381},
      {'label': 'disaster', 'score': 0.9028499126434326},
      {'label': 'general', 'score': 0.8125467896461487},
      {'label': 'disaster', 'score': 0.9024186134338379},
      {'label': 'general', 'score': 0.7888662219047546},
      {'label': 'disaster', 'score': 0.8984711170196533},
      {'label': 'general', 'score': 0.8884401917457581}]

```

```

[12]: import pandas as pd
data = pd.DataFrame(df)
data
data['Review'] = text
data['Predicted Sentiment Score'] = data['score']*100
data['Predicted Label'] = data['label']
data = data.drop(['label', 'score'], axis = 1)
data

```

```

[12]:                                     Review \
0  BREAKING: A massive 6.8 magnitude earthquake j...
1  My laptop just crashed right before I saved my...
2  The wildfire is spreading rapidly due to high ...
3  That new Taylor Swift track is absolute fire! ...
4  Flash flood warnings active for downtown. Mult...
5  We are drowning in paperwork at the office tod...
6  Emergency crews are responding to a structural...

```

```

7 I am burning up in this summer heat! The AC at...
8 A major multi-vehicle accident has completely ...
9 My favorite character just got killed off in t...

```

	Predicted Sentiment Score	Predicted Label
0	90.274811	disaster
1	71.864343	general
2	90.238255	disaster
3	83.904815	general
4	90.284991	disaster
5	81.254679	general
6	90.241861	disaster
7	78.886622	general
8	89.847112	disaster
9	88.844019	general

```
[13]: data.to_csv('tinybert_sentiment_results.csv', index=False)
```

6.1 Push Model to AWS S3

```
[ ]: import boto3

s3 = boto3.client('s3')

bucket_name = 'disaster-classification-model-tinybert'

def create_bucket(bucket_name):
    response = s3.list_buckets()
    buckets = [buck['Name'] for buck in response['Buckets']]
    if bucket_name not in buckets:
        s3.create_bucket(Bucket=bucket_name)
        print("Bucket is Created")
    else:
        print("Bucket already exists in your account!!! Fell free to use it..")

create_bucket(bucket_name)
```

6.2 Uploading the Model Folder to S3 Bucket disaster-classification-model-tinybert/tinybert-disaster-tweet

```
[ ]: import os

import boto3
s3 = boto3.client('s3')

bucket_name = 'disaster-classification-model-tinybert'
```

```

def upload_directory(directory_path, s3_prefix):
    for root, dirs, files in os.walk(directory_path):
        for file in files:
            file_path = os.path.join(root, file).replace("\\", "/")
            relpath = os.path.relpath(file_path, directory_path)
            s3_key = os.path.join(s3_prefix, relpath).replace("\\", "/")

            s3.upload_file(file_path, bucket_name, s3_key)

upload_directory('tinybert-disaster-tweet',
    ↪ 'disaster-classification-model-tinybert/tinybert-disaster-tweet')

```